

# **Home Lab 01: Perform System Configuration Gap Analysis**

## **1. Lab Overview**

In this lab, I performed a **system configuration gap analysis** on a Windows Server 2019 virtual machine named **PC10**. The goal was to compare the system's current security configuration against a Microsoft security baseline template.

This lab helped me understand how security and IT teams identify misconfigurations, compare systems against recommended standards, and determine whether a system is compliant with a required security baseline. The lab used **Microsoft Policy Analyzer** and the **Windows Server 2019 Security Baseline** from the Microsoft Security Compliance Toolkit.

## **2. Business Scenario**

As a security team member at **Structureality Inc.**, I was tasked with helping improve the organization's security posture. A key part of securing an IT environment is comparing the current system configuration against a trusted security baseline.

To complete this task, I used Microsoft Policy Analyzer to review the current configuration of a Windows Server 2019 system and identify gaps between the system's actual settings and the recommended baseline settings.

## **3. Lab Objective**

The objective of this lab was to:

- Perform a system configuration gap analysis.
- Compare a Windows Server 2019 system against a security baseline.
- Identify differences between the baseline configuration and the current effective system state.
- Understand how misconfigurations can affect security.
- Practice using Microsoft Policy Analyzer for baseline review.

A gap analysis compares the current configuration of a system with a template, baseline, framework, or settings document to identify differences between the expected configuration and the actual configuration.

## **4. Lab Environment**

| Item              | Details                                                           |
|-------------------|-------------------------------------------------------------------|
| Virtual Machine   | PC10                                                              |
| Operating System  | Windows Server 2019                                               |
| Windows Version   | 1809                                                              |
| OS Build          | 17763.4377                                                        |
| Tool Used         | Microsoft Policy Analyzer                                         |
| Security Baseline | Windows 10 Version 1809 and Windows Server 2019 Security Baseline |
| Source Toolkit    | Microsoft Security Compliance Toolkit                             |



Figure 1: Windows Server 2019 version and OS build verified using winver.

## 5. Tools Used

- Windows Server 2019
- Microsoft Policy Analyzer
- Microsoft Security Compliance Toolkit
- Windows Server 2019 Security Baseline
- PowerShell
- File Explorer
- Virtual lab environment

## 6. Commands Used

The following commands were used during the lab:

```
copy D:\* C:\LABFILES
```

This command copied the Policy Analyzer and Windows Server 2019 security baseline files from the lab media to the local lab folder.

```
cd C:\LABFILES
```

This command changed the working directory to the lab files folder.

```
ls
```

This command displayed the contents of the lab folder.

```
Expand-Archive -Path PolicyAnalyzer.zip
```

This command extracted the Policy Analyzer files.

```
Expand-Archive -Path "Windows 10 Version 1809 and Windows Server 2019 Security Baseline.zip"
```

This command extracted the Windows Server 2019 security baseline files.

```
C:\LABFILES\PolicyAnalyzer\PolicyAnalyzer_40\PolicyAnalyzer.exe
```

This command launched the Microsoft Policy Analyzer application.

Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.

```
PS C:\Windows\system32> copy D:\* c:\LABFILES
PS C:\Windows\system32> cd c:\LABFILES
PS C:\LABFILES> ls
```

Directory: C:\LABFILES

| Mode   | LastWriteTime      | Length  | Name                                                                  |
|--------|--------------------|---------|-----------------------------------------------------------------------|
| d----  | 10/22/2018 7:57 PM |         | contains-nothing                                                      |
| d----  | 10/22/2018 7:57 PM |         | empty                                                                 |
| d----  | 6/5/2023 1:48 AM   |         | MARKETING                                                             |
| d----  | 6/5/2023 1:48 AM   |         | NVD-Control-RA-5-VULNERABILITY_SCANNING_files                         |
| d----  | 6/5/2023 1:48 AM   |         | pcaps                                                                 |
| d----  | 6/5/2023 1:48 AM   |         | ScoutSuite                                                            |
| d----  | 6/5/2023 1:48 AM   |         | winlogbeat                                                            |
| -a---- | 3/11/2021 11:13 AM | 7520    | 515tech_store.sql                                                     |
| -a---- | 8/22/2018 7:44 AM  | 54158   | comptia-logo.jpg                                                      |
| -a---- | 3/28/2019 6:31 AM  | 24      | CONFIDENTIAL.txt                                                      |
| -a---- | 3/9/2020 7:00 AM   | 158621  | conn-sample.log                                                       |
| -a---- | 2/1/2020 1:37 PM   | 16298   | CSIRT Incident Handling Form.docx                                     |
| -a---- | 2/1/2020 1:39 PM   | 106194  | CSIRT Incident Handling Form.pdf                                      |
| -a---- | 3/22/2021 2:30 PM  | 257     | DisableController.ps1                                                 |
| -a---- | 3/22/2021 11:40 AM | 457     | DisableDisk.ps1                                                       |
| -a---- | 6/28/2022 12:58 AM | 1883    | iam_shares.ps1                                                        |
| -a---- | 2/21/2023 3:44 AM  | 328024  | laptop-full.pcapng                                                    |
| -a---- | 2/21/2023 3:47 AM  | 312488  | laptop-selected.pcapng                                                |
| -a---- | 3/10/2020 5:26 AM  | 496     | local.rules                                                           |
| -a---- | 1/21/2020 4:29 AM  | 71966   | NVD-Control-RA-5-VULNERABILITY_SCANNING.html                          |
| -ar--- | 6/18/2023 10:06 AM | 1592778 | PolicyAnalyzer.zip                                                    |
| -a---- | 3/11/2021 1:32 AM  | 233     | set_default_password.ps1                                              |
| -a---- | 6/28/2022 1:08 AM  | 44725   | Structureality-netdiag.odg                                            |
| -a---- | 6/29/2022 3:56 AM  | 560     | trusted-installs.csv                                                  |
| -ar--- | 6/18/2023 10:07 AM | 1395988 | Windows 10 Version 1809 and Windows Server 2019 Security Baseline.zip |

```
Administrator: Windows PowerShell
d----- 10/22/2018 7:57 PM contains-nothing
d----- 10/22/2018 7:57 PM empty
d----- 6/5/2023 1:48 AM MARKETING
d----- 6/5/2023 1:48 AM NVD-Control-RA-5-VULNERABILITY_SCANNING_files
d----- 6/5/2023 1:48 AM pcaps
d----- 6/5/2023 1:48 AM ScoutSuite
d----- 6/5/2023 1:48 AM winlogbeat
-a----- 3/11/2021 11:13 AM 7520 51Stech_store.sql
-a----- 8/22/2018 7:44 AM 54158 comptia-logo.jpg
-a----- 3/28/2019 6:31 AM 24 CONFIDENTIAL.txt
-a----- 3/9/2020 7:00 AM 158621 conn-sample.log
-a----- 2/1/2020 1:37 PM 16298 CSIRT Incident Handling Form.docx
-a----- 2/1/2020 1:39 PM 106194 CSIRT Incident Handling Form.pdf
-a----- 3/22/2021 2:30 PM 257 DisableController.ps1
-a----- 3/22/2021 11:40 AM 457 DisableDisk.ps1
-a----- 6/28/2022 12:58 AM 1883 iam_shares.ps1
-a----- 2/21/2023 3:44 AM 328024 laptop-full.pcapng
-a----- 2/21/2023 3:47 AM 312488 laptop-selected.pcapng
-a----- 3/10/2020 5:26 AM 496 local.rules
-a----- 1/21/2020 4:29 AM 71966 NVD-Control-RA-5-VULNERABILITY_SCANNING.html
-ar----- 6/18/2023 10:06 AM 1592778 PolicyAnalyzer.zip
-a----- 3/11/2021 1:32 AM 233 set_default_password.ps1
-a----- 6/28/2022 1:08 AM 44725 Structureality-netdiag.odg
-a----- 6/29/2022 3:56 AM 560 trusted-installs.csv
-ar----- 6/18/2023 10:07 AM 1395988 Windows 10 Version 1809 and Windows Server 2019 Security Baseline.zip
ine.zip

PS C:\LABFILES> Expand-Archive -Path PolicyAnalyzer.zip
PS C:\LABFILES> Expand-Archive -Path "Windows 10 Version 1809 and Windows Server 2019 Security Baseline.zip"
PS C:\LABFILES> C:\LABFILES\PolicyAnalyzer\PolicyAnalyzer_40\PolicyAnalyzer.exe
PS C:\LABFILES>
```

Figure 2: PowerShell commands used to copy and verify the Policy Analyzer and security baseline files.

## 7. Steps Performed

### Step 1: Verified Windows Server Version

I first verified the Windows Server version and build number using `winver`. This was important because the correct security baseline had to match the operating system version.

#### Result:

The system was running **Windows Server 2019 Version 1809**, OS Build **17763.4377**.

### Step 2: Copied Lab Files

I copied the Policy Analyzer and Windows Server 2019 security baseline files from the removable lab media to `C:\LABFILES`.

This prepared the system for baseline comparison.

### **Step 3: Extracted Policy Analyzer and Security Baseline Files**

I extracted both compressed files:

- `PolicyAnalyzer.zip`
- `Windows 10 Version 1809 and Windows Server 2019 Security Baseline.zip`

After extraction, the files were ready to be used in Microsoft Policy Analyzer.

### **Step 4: Opened Microsoft Policy Analyzer**

I launched Policy Analyzer from the extracted folder.

Policy Analyzer allowed me to view, compare, and analyze the security baseline settings.

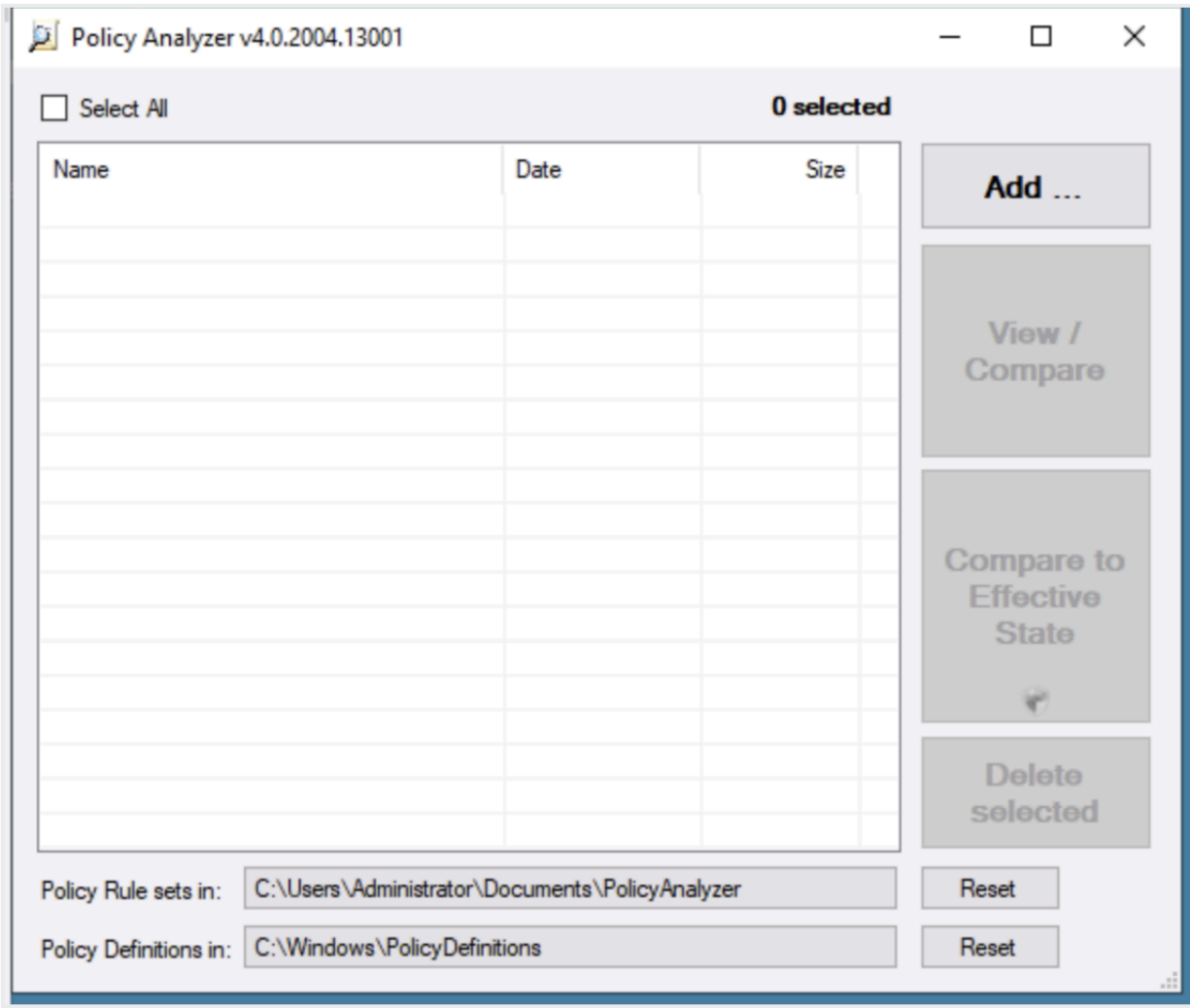


Figure 3: Microsoft Policy Analyzer opened to begin the security baseline review.

### Step 5: Loaded the Security Baseline

Inside Policy Analyzer, I selected the folder containing the Policy Analyzer policy rules files. The baseline rule sets appeared in the Policy Analyzer window.



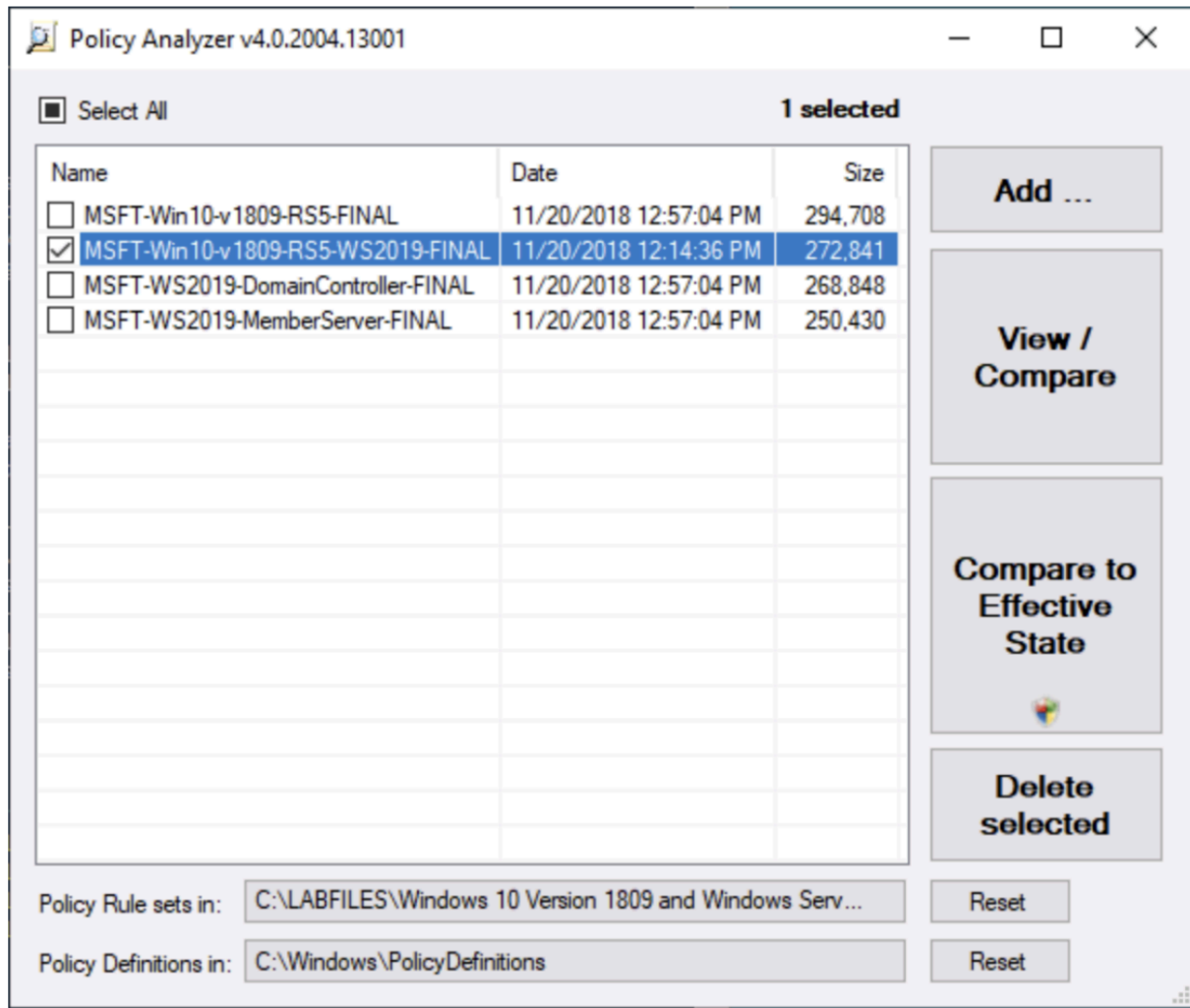


Figure 4: Windows Server 2019 security baseline rule sets loaded in Microsoft Policy Analyzer.

The selected baseline was:

MSFT-Win10-v1809-RS5-WS2019-FINAL

This baseline contained the recommended security settings for Windows 10 Version 1809 and Windows Server 2019.

## Step 6: Viewed the Baseline Settings

I selected **View / Compare** to review the settings inside the baseline.

This allowed me to see the recommended configuration values from the security template.

| Policy Viewer - 393 items     |                              |                                 |                 |
|-------------------------------|------------------------------|---------------------------------|-----------------|
| Clipboard View Export Options |                              |                                 |                 |
| Policy Type                   | Policy Group or Registry Key | Policy Setting                  | MSFT-Win10-v180 |
| Security Template             | Privilege Rights             | SeSecurityPrivilege             | *S-1-5-32-544   |
| Security Template             | Privilege Rights             | SeSystemEnvironmentPrivilege    | *S-1-5-32-544   |
| Security Template             | Privilege Rights             | SeTakeOwnershipPrivilege        | *S-1-5-32-544   |
| Security Template             | Privilege Rights             | SeTcbPrivilege                  |                 |
| Security Template             | Privilege Rights             | SeTrustedCredManAccessPrivilege |                 |
| Security Template             | Service General Setting      | "AppIDSvc"                      | 2 ""            |
| Security Template             | Service General Setting      | "XblAuthManager"                | 4 ""            |
| Security Template             | Service General Setting      | "XblGameSave"                   | 4 ""            |
| Security Template             | Service General Setting      | "XboxGipSvc"                    | 4 ""            |
| Security Template             | Service General Setting      | "XboxNetApiSvc"                 | 4 ""            |
| Security Template             | System Access                | ClearTextPassword               | 0               |
| Security Template             | System Access                | EnableAdminAccount              | 0               |
| Security Template             | System Access                | EnableGuestAccount              | 0               |
| Security Template             | System Access                | LockoutBadCount                 | 10              |
| Security Template             | System Access                | LockoutDuration                 | 15              |
| Security Template             | System Access                | LSAAnonymousNameLookup          | 0               |
| Security Template             | System Access                | MaximumPasswordAge              | 60              |
| Security Template             | System Access                | MinimumPasswordAge              | 1               |
| Security Template             | System Access                | MinimumPasswordLength           | 14              |
| Security Template             | System Access                | PasswordComplexity              | 1               |
| Security Template             | System Access                | PasswordHistorySize             | 24              |
| Security Template             | System Access                | ResetLockoutCount               | 15              |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Policy Path:</b><br>Advanced Audit Policy Configuration<br>System Audit Policies\Account Logon<br>Credential Validation<br><br><i>Credential Validation</i><br><br><i>This policy setting allows you to audit events generated by validation tests on user account logon credentials.</i><br><br><i>Events in this subcategory occur only on the computer that is authoritative for those credentials. For domain accounts, the domain controller is authoritative. For local accounts, the local computer is authoritative.</i><br><br><i>Volume: High on domain controllers.</i><br><br><i>Default on Client editions: No Auditing.</i><br><br><i>Default on Server editions: Success.</i> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------|
| <b>MSFT-Win10-v1809-RS5-WS2019-FINAL:</b><br><i>Option:</i> Success and Failure<br><b>Defined in the following GPOs:</b> |
|--------------------------------------------------------------------------------------------------------------------------|

Figure 5: Baseline security policy settings displayed in Policy Analyzer.

Two important baseline values identified were:

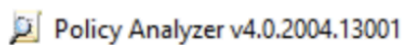
| Policy Setting        | Baseline Value |
|-----------------------|----------------|
| LockoutBadCount       | 10             |
| MinimumPasswordLength | 14             |

The baseline value for **LockoutBadCount** was **10**, and the baseline value for **MinimumPasswordLength** was **14**.

## **Step 7: Compared Baseline to Effective State**

Next, I selected **Compare to Effective State**.

This compared the recommended baseline settings against the current live configuration of the PC10 system.

☐ Select All

2 selected

[illegible]

Add ...

View / Compare

### Compare to Effective State

Delete  
selected

Policy Rule sets in: C:\LABFILES\Windows 10 Version 1809 and Windows Serv...

Reset

Policy Definitions in: C:\Windows\PolicyDefinitions

Reset

| Policy Viewer - 393 items     |                              |                                 |                     |                     |
|-------------------------------|------------------------------|---------------------------------|---------------------|---------------------|
| Clipboard View Export Options |                              |                                 |                     |                     |
| Policy Type                   | Policy Group or Registry Key | Policy Setting                  | Baseline(s)         | Effective state     |
| Audit Policy                  | Account Logon                | Credential Validation           | Success and Fail... | Success             |
| Audit Policy                  | Account Management           | Computer Account Management     | Success             | Success             |
| Audit Policy                  | Account Management           | Other Account Management Events | Success             | No Auditing         |
| Audit Policy                  | Account Management           | Security Group Management       | Success             | Success             |
| Audit Policy                  | Account Management           | User Account Management         | Success and Fail... | Success             |
| Audit Policy                  | Detailed Tracking            | PNP Activity                    | Success             | No Auditing         |
| Audit Policy                  | Detailed Tracking            | Process Creation                | Success             | No Auditing         |
| Audit Policy                  | DS Access                    | Directory Service Access        | Success and Fail... | Success             |
| Audit Policy                  | DS Access                    | Directory Service Changes       | Success and Fail... | No Auditing         |
| Audit Policy                  | Logon/Logoff                 | Account Lockout                 | Failure             | Success             |
| Audit Policy                  | Logon/Logoff                 | Group Membership                | Success             | No Auditing         |
| Audit Policy                  | Logon/Logoff                 | Logon                           | Success and Fail... | Success and Fail... |
| Audit Policy                  | Logon/Logoff                 | Other Logon/Logoff Events       | Success and Fail... | No Auditing         |
| Audit Policy                  | Logon/Logoff                 | Special Logon                   | Success             | Success             |
| Audit Policy                  | Object Access                | Detailed File Share             | Failure             | No Auditing         |
| Audit Policy                  | Object Access                | File Share                      | Success and Fail... | No Auditing         |
| Audit Policy                  | Object Access                | Other Object Access Events      | Success and Fail... | No Auditing         |
| Audit Policy                  | Object Access                | Removable Storage               | Success and Fail... | No Auditing         |
| Audit Policy                  | Policy Change                | Audit Policy Change             | Success             | Success             |
| Audit Policy                  | Policy Change                | Authentication Policy Change    | Success             | Success             |
| Audit Policy                  | Policy Change                | MPSSVC Rule-Level Policy Change | Success and Fail... | No Auditing         |
| Audit Policy                  | Policy Change                | Other Policy Change Events      | Failure             | No Auditing         |
| Audit Policy                  | Privilege Use                | Sensitive Privilege Use         | Success and Fail... | No Auditing         |

**Policy Path:**  
Advanced Audit Policy Configuration  
System Audit Policies\Account Logon  
Credential Validation  
  
**Credential Validation**  
This policy setting allows you to audit events generated by validation tests on user account logon credentials.  
Events in this subcategory occur only on the computer that is authoritative for those credentials. For domain accounts, the domain controller is authoritative. For local accounts, the local computer is authoritative.  
Volume: High on domain controllers.  
Default on Client editions: No Auditing.  
Default on Server editions: Success.  
  
**Baseline(s):**  
Option: Success and Failure  
Defined in the following GPOs:

Figure 6-7: Gap analysis showing differences between the baseline and the current effective system state.

The comparison showed many highlighted items, which indicated differences between the baseline file and the system's current effective state.

## 8. Key Findings

The gap analysis found that the PC10 system was **not fully compliant** with the security baseline.

| Policy Setting | Baseline Value | Effective System Value | Status |
|----------------|----------------|------------------------|--------|
|----------------|----------------|------------------------|--------|

|                       |    |   |               |
|-----------------------|----|---|---------------|
| LockoutBadCount       | 10 | 0 | Not Compliant |
| MinimumPasswordLength | 14 | 7 | Not Compliant |

The effective state value for **LockoutBadCount** was **0**, while the baseline expected **10**. The effective state value for **MinimumPasswordLength** was **7**, while the baseline expected **14**. Based on these results, PC10 was not compliant with the security template.

|                        |    |    |
|------------------------|----|----|
| ClearTextPassword      | 0  | 0  |
| EnableAdminAccount     | 0  | 1  |
| EnableGuestAccount     | 0  | 0  |
| LockoutBadCount        | 10 | 0  |
| LockoutDuration        | 15 |    |
| LSAAnonymousNameLookup | 0  | 0  |
| MaximumPasswordAge     | 60 | 42 |
| MinimumPasswordAge     | 1  | 1  |
| MinimumPasswordLength  | 14 | 7  |
| PasswordComplexity     | 1  | 1  |
| PasswordHistorySize    | 24 | 24 |
| ResetLockoutCount      | 15 |    |

Figure 8: Non-compliant password and account lockout settings identified during the gap analysis.

## 9. Analysis

The results showed that the system's current password and account lockout settings were weaker than the recommended baseline.

The most important findings were:

- The system did not enforce the recommended minimum password length.

- The system did not match the recommended account lockout threshold.
- Several security settings differed from the expected baseline.
- The system required configuration changes before it could be considered compliant.

This matters because weak password and lockout settings can increase the risk of unauthorized access, password guessing, and account compromise.

## 10. Recommendation

Based on the gap analysis, I would recommend the following actions:

1. Update the system's password policy to match the baseline requirement.
2. Set the minimum password length to **14 characters**.
3. Configure the account lockout threshold to match the recommended baseline.
4. Review all highlighted differences in Policy Analyzer.
5. Apply only approved changes through Group Policy or local security policy.
6. Re-run the gap analysis after making changes to confirm compliance.
7. Document all configuration changes for audit and compliance purposes.

## 11. IT Support Relevance

This lab is useful for an IT Support Analyst because support teams often help maintain secure and properly configured systems.

This lab connects to real IT support tasks such as:

- Checking Windows system settings
- Reviewing password policy issues
- Understanding account lockout settings
- Identifying system misconfigurations
- Supporting compliance requirements
- Using administrative tools
- Documenting technical findings
- Escalating security configuration issues when needed

For an entry-level IT support role, this lab shows that I understand how to compare a system's current settings against company security standards and identify areas that need correction.

## 12. Skills Demonstrated

This lab demonstrates the following skills:

- Windows Server 2019 administration
- Security baseline analysis
- Configuration gap analysis
- Microsoft Policy Analyzer usage
- PowerShell basics
- Password policy review
- Account lockout policy review
- Compliance checking
- Technical documentation
- Security-focused troubleshooting

### **13. Ticket-Style Summary**

**Issue:**

PC10 needed to be reviewed for compliance against a Windows Server 2019 security baseline.

**Impact:**

The system had security settings that did not match the recommended baseline, which could increase security risk.

**Troubleshooting Steps:**

- Verified the Windows Server version and build number.
- Copied the required baseline and Policy Analyzer files to the local system.
- Extracted the baseline and Policy Analyzer files.
- Opened Microsoft Policy Analyzer.
- Loaded the Windows Server 2019 security baseline.
- Viewed the baseline configuration.
- Compared the baseline against the system's effective state.
- Identified non-compliant settings.

**Root Cause:**

The current effective system settings did not match the recommended security baseline values.

**Resolution/Recommendation:**

Update the non-compliant password and account lockout settings to match the approved security baseline, then perform another gap analysis to confirm compliance.



**Status:**

Analysis completed. Remediation recommended.

## **14. Lessons Learned**

**In this lab, I learned that a system can appear to be working normally but still fail to meet security standards. I also learned that gap analysis does not automatically fix the system. It identifies the differences between the current configuration and the expected configuration.**

**This helped me understand why security baselines are important in business environments. They give IT teams a standard to compare systems against and help identify weak configurations before they become security problems.**